

BYU Civil Engineering Program Assessment Manual

CE Undergraduate Committee

2025-04-01

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Introduction

This document describes the processes and methods we use to assess and evaluate the BYU Civil Engineering (BS) program. ABET criterion 4 states:

The program must regularly use appropriate, documented processes for assessing and evaluating the extent to which the student outcomes are being attained. The results of these evaluations must be systematically utilized as input for the program's continuous improvement actions. Other available information may also be used to assist in the continuous improvement of the program.

This manual documents these processes and provides a record of historical assessment data. The program uses five categories of assessments:

1. Direct assessment of student outcomes in course activities (see Chapter 1)
2. Graduation surveys and exit interviews (see Chapter 2)
3. Student performance on the Fundamentals of Engineering (FE) exam (see Chapter 3)
4. Student course evaluations (see Chapter 4)
5. Statistics on student progress in the major (see Chapter 5)

Evaluation and Continuous Improvement Process

Each summer, the CE undergraduate committee evaluates recent assessment data. The evaluation process results in a determination of the program's strengths and weaknesses. The process also recommends actions to resolve the identified weaknesses. Potential actions may include changes to the following:

- program courses, requirements, or prerequisites
- course content or learning activities
- course assessments
- performance indicators or associated rubrics

The undergraduate committee documents the assessment evidence, evaluated strengths and weaknesses, and proposed actions in the program stewardship report. This report is reviewed by the department faculty and approved in a meeting before August 31 of the year of the report.

The undergraduate committee takes responsibility to implement the recommended actions. Changes to program courses, requirements, or prerequisites are approved by faculty vote and submitted to the university curriculum committee and registrar for approval. Changes to course learning activities are recommended to course faculty, who prepare a summary report of their actions.

Instructors regularly make improvements to course activities, pedagogy, and content based on data obtained through direct experience and student evaluation comments. These are important data that may lead to recommended course and program changes, but are not a formal assessment tool.

Chapter 1

Student Outcomes

The program has seven student outcomes, which describe abilities and skills that students should demonstrate upon graduation.

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

The primary assessment of these student outcomes is a direct assessment of individual student work undertaken at multiple points in the program curriculum. We attempt to formally assess each student outcome at two levels:

1. At a Formative level, in lower-division courses. There may be more than one formative assessment of a student outcome.
2. At a Summative level, in an upper-division course as close to graduation as possible.

Summative assessments are given priority in program evaluation, but formative assessments can provide additional supportive data. Table 1.1 lists which courses are used for formative and summative assessments. Formative assessments may not assess all indicators of a student outcome, but summative assessments cover all indicators. Additionally, formative assessments may include data collected from students in other programs.

We assess each outcome on a three-year schedule. Table 1.2 shows two cycles of the three-year assessment schedule. In a six-year ABET accreditation cycle, we collect data on student performance for each outcome twice. Courses are assessed in one of Fall or Winter semesters for courses that are offered in both semesters. Early in the assessment semester, instructors are informed that they will be expected to provide assessment data in their class. At the end of the semester, instructors are invited to complete a form that requires a brief description of the assessment (e.g., final exam question), a specific assessment instrument as given to students, and the number of students assessed at each level of the rubric.

Table 1.1: Student Outcome Assessment Locations

Outcome	CCE102	CCE103	Software	Measurements	CCE203	CE232	CCE270	CE291	Breadth	Design	Capstone
1. Complex problems		F			F				S		
2. Design										F	S
3. Communication				F			F				S
4. Ethics		F						F			S
5. Teamwork		F						F			S
6. Experiments and data						F	F		S ^a		
7. Acquire knowledge			F					F			S

^a Only selected breadth courses contribute to the summative assessment.

Table 1.2: Assessment Schedule

Outcome	25 Winter	25 Sp/Su	25 Fall	26 Winter	26 Sp/Su	26 Fall	27 Winter	27 Sp/Su	27 Fall	28 Winter	28 Sp/Su
SO1: Complex Problems						A	E		C		
SO2: Design			A	A	E	C					
SO3: Communications						A	E		C		
SO4: Ethics	A	E	C							A	E
SO5: Teamwork			A	A	E	C					
SO6: Experiments / Data	A	E	C							A	E
SO7: Acquire Knowledge			A	A	E	C					
ABET Schedule					Self-study	Visit	Response				

A: Assessment, E: Evaluation, C: Change

For each of the seven student outcomes, the program develops a set of performance indicators and an associated rubric that faculty apply to specific and selected student assessments in the assessed courses. The levels of the rubric are

- Unsatisfactory
- Developing
- Meets Expectations
- Exceeds Expectations

The performance indicators and rubric apply to all levels of instruction, though the level of expected student performance increases as students grow and learn through the program.

For all indicators, the program goal is that 75 percent of students will demonstrate performance that meets or exceeds expectations by graduation. The following sections describe the performance indicators and associated rubrics used to assess each student outcome, along with a brief description of the assessment instruments used for each course and assessment and evaluation information. Specific assessment instruments are archived and available in the program assessment folder.

The annual program stewardship report includes the present academic year's assessment and evaluation, as well as reports on actions taken related to the previous year's recommended changes.

Assessment Practices

The program assessment coordinator creates a Microsoft Forms survey for each student outcome. In the semester that a course is assessed, the program assessment coordinator sends the survey to the instructor at least twice:

1. Before the semester begins, to remind the instructor of their assessment responsibilities and to invite them to consider activities that will be used for the assessment.
2. At the end of the semester, to collect the final assessment data.

The survey asks instructors to provide **for each indicator**:

- A brief description of the indicator; e.g., "Final exam question"
- The assessment instrument as given to students; e.g. the text of the final exam question with accompanying diagrams or data. This must be provided as a PDF file.
- The number of students assessed at each level of the rubric for the indicator.

The assessment coordinator collects the assessment data from the Forms databases and transfers them into this repository. The coordinator cleans the data as necessary to create the assessment figures and commits the rendered page.

Best Practices for Instructors

Activities used for student outcomes assessment should be clearly labeled on the syllabus; i.e., "HW 4.1. This problem is used to assess student outcome X in our program assessments." This accomplishes two things. First, it helps instructors to understand what activities are used in assessment as the course is taught by others. Second, it communicates to the students that their performance on the activity is used for more than their own grade.

A single activity may be used to assess multiple student outcomes, or multiple indicators of a single outcome. But each indicator should be assessed separately based on the rubric for that indicator. For example, a term project may provide assessment data on both "Communication: Write" indicator and on the "Design: Recommend design" indicators. But student performance on the indicator should be assessed separately, based on the rubric.

It is not required that the rubrics listed in this section be determinative of the grade the students receive on the activity. That is, an instructor may still elect to issue full marks for work assessed at a

“Developing” level, as an example. However, it is best practice to conform the rubric used in grading to the rubric used in the assessment, so that students are fully informed of our expectations for their performance. In the term project example above, an instructor may choose to award more points for the “Communication: Write” indicator than for the “Design: Recommend design” indicator based on their goals for the course.

Instructors should select work that students complete **independently** if at all possible. Team project reports or other collaborative efforts make it difficult to assess student performance. When assessing teamwork-related indicators using team projects, instructors should make an effort to understand how individual students performed.

Learning Suite contains tools that allow instructors to link assignments to course learning outcomes. Doing this automates many features of collecting assessment data.

1.1 Complex Problems

Students will demonstrate:

An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

ABET Definition

The ABET definition of “Complex Problems” is

Complex engineering problems include one or more of the following characteristics: involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.

It is understood that just because a problem is “hard” does not mean it is complex. Similarly, a problem may have many sub-steps, but if the parts are not integrated or coordinated the problem would not be considered “complex.”

1.1.1 Performance Indicators

We evaluate this outcome based on three performance indicators. Students should:

1. Identify the problem parameters and given values.
2. Formulate a solution strategy.
3. Evaluate the best alternative from many possible solutions.

We assess these performance indicators with the rubric in Table 1.3.

Table 1.3: Student Outcome 1 Assessment Rubric

Indicator	Unsatisfactory	Developing	Meets Expectations	Exceeds Expectations
1. Identify	Fails to identify given parameters.	Identifies some given parameters.	Identifies all provided parameters.	Identifies all parameters including unreasonable parameters or those requiring assumptions.
2. Formulate	Fails to formulate a solution strategy.	Formulates a strategy with limitations.	Formulates a solution strategy based in accepted methods.	Formulates a strategy that adjusts for the weaknesses of existing techniques.
3. Evaluate	Fails to evaluate alternative solutions.	Evaluates an alternative solution.	Evaluates alternative solutions.	Evaluates alternatives with a consideration of human needs.

1.1.2 Assessment Instruments

We assess this student outcome in the following courses with the listed assessments:

Formative

- CCE 103: Engineering Mechanics – Statics
 - Identify: Final exam question
 - Formulate: Final exam question
- CCE 203: Engineering Mechanics – Materials
 - Identify: Final exam question
 - Formulate: Final exam question

Summative

- CE 321: Structural Analysis
 - Identify: Truss analysis project
 - Formulate: Truss analysis project
 - Evaluate: Truss analysis project
- CE 331: Hydrology
 - Identify: Term project
 - Formulate: Term project
 - Evaluate: Term project
- CE 341: Soil Mechanics
 - Identify: Final exam question
 - Formulate: Final exam question
 - Evaluate: Final exam question
- CE 351: Environmental Engineering
 - Identify: Final exam question
 - Formulate: Final exam question
 - Evaluate: Final exam question
- CE 361: Transportation Engineering
 - Identify: Pavement management homework
 - Formulate: Pavement management homework
 - Evaluate: Pavement management homework

1.1.3 Evaluation

We will assess and evaluate this student outcome in the 2026-27 academic year.

1.2 Design and Contemporary Issues

Students will demonstrate:

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare as well as global, cultural, social, environmental, and economic factors.

ABET Definition

The ABET definition of “Design” is

Engineering design is a process of devising a system, component, or process to meet desired needs and specifications within constraints. It is an iterative, creative, decision-making process in which the basic sciences, mathematics, and engineering sciences are applied to convert resources into solutions. Engineering design involves identifying opportunities, developing requirements, performing analysis and synthesis, generating multiple solutions, evaluating solutions against requirements, considering risks, and making trade-offs, for the purpose of obtaining a high-quality solution under the given circumstances. For illustrative purposes only, examples of possible constraints include aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability.

1.2.1 Performance Indicators

We evaluate this outcome based on four performance indicators. Students should:

1. Develop the criteria against which the design will be judged. These criteria may include safety requirements, performance expectations, or industry standards.
2. Select appropriate computational, experimental, or qualitative methods to develop the design.
3. Justify the selection of a preferred solution from among multiple alternatives.
4. Consider public health, safety, and welfare as well as global, cultural, social, environmental, and economic factors.

We assess these performance indicators with the rubric in Table 1.4.

Table 1.4: Student Outcome 2 Assessment Rubric

Indicator	Unsatisfactory	Developing	Meets Expectations	Exceeds Expectations
1. Develop design criteria	Little recognition of criteria or selection of irrelevant criteria.	Partial recognition of relevant criteria.	Thorough recognition of relevant criteria.	Thorough recognition of relevant criteria, plus optional ones.
2. Select methods	Poor matching of methods to problem.	Partial selection or justification of relevant methods.	Selection and justification of methods relevant to solving problem.	Selection and justification of relevant methods, acknowledging limitations.

Table 1.5: Assessment Instruments for Student Outcome 2

Assessment Level	Course	Instrument
Develop design criteria		
Formative	CE 421	"Final exam question"
Formative	CE 424	Question from Midterm Exam 2
Summative	CE 472	Final report
Recommend design		
Formative	CE 421	Final Exam Question
Formative	CE 424	Question from MT1
Formative	CE 433	Canal design for Heber City
Formative	CE 453	HW No. 4: Mixing and Clarification Design
Formative	CE 461	Final design project
Summative	CE 472	Final report
Select methods		
Formative	CE 421	Final Exam Question
Summative	CE 472	Final report
Consider Factors		
Formative	CE 421	Final Exam Question
Summative	CE 472	Final report

Indicator	Unsatisfactory	Developing	Meets Expectations	Exceeds Expectations
3. Recommend design	No design, wrong design, or unjustified design.	Partially acceptable design and justification.	Acceptable design justified by methods and criteria.	Acceptable, justified design chosen from multiple alternatives.
4. Consider factors	Unsatisfactory		Meets Expectations	
Public Health	Does not consider		Considers	
Safety	Does not consider		Considers	
Welfare	Does not consider		Considers	
Global	Does not consider		Considers	
Cultural	Does not consider		Considers	
Social	Does not consider		Considers	
Environmental	Does not consider		Considers	
Economic	Does not consider		Considers	

1.2.2 Assessment Instruments

We assess this student outcome in the following courses with the assessments listed in Table 1.5. Complete instrument prompts are available in the program assessment folder.

1.2.3 Data and Evaluation

Figure 1.1 shows the share of students assessed at each performance level for each indicator in the relevant courses. The assessment is as follows:

- **Develop design criteria:** Two formative assessments (CE 421 and CE 424) and one summative assessment (CE 472) reveal that program students are meeting expectations at the 75% threshold level.
- **Select methods:** One formative assessment (CE 421) and one summative assessment (CE 472) both assess 100% of students at the meeting expectations level.
- **Recommend design:** Five formative assessments and one summative assessment (CE 472) vary in their assessment of program students. In CE 421 and 433, students are meeting expectations at the 75% threshold level. In the other courses — especially in the summative assessment of CE 472 — students are not meeting expectations at the 75% threshold.

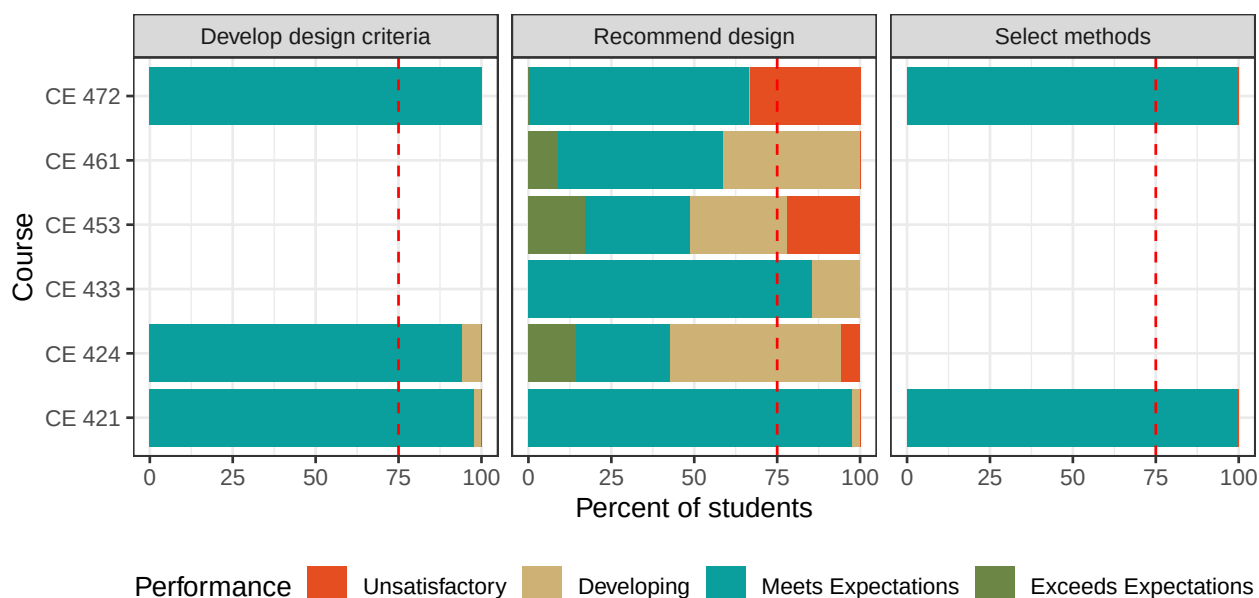


Figure 1.1: Fall 2025 and Winter 2026 Assessment of Student Outcome 2

Figure 1.2 shows the share of students who consider the various design factors in the “Consider factors” performance indicator. In the formative assessment of CE 421, all students were assessed as considering the factors appropriately. In CE 472, students met the 75% expectations threshold for Safety, Environmental, and Economic factors but not for the Public Health, Welfare, Global, Cultural, or Social factors.

The committee recommends that more design courses develop activities that will instruct students in topics related to “Develop design criteria,” “Select methods,” and “Consider factors” indicators and assess the activities.

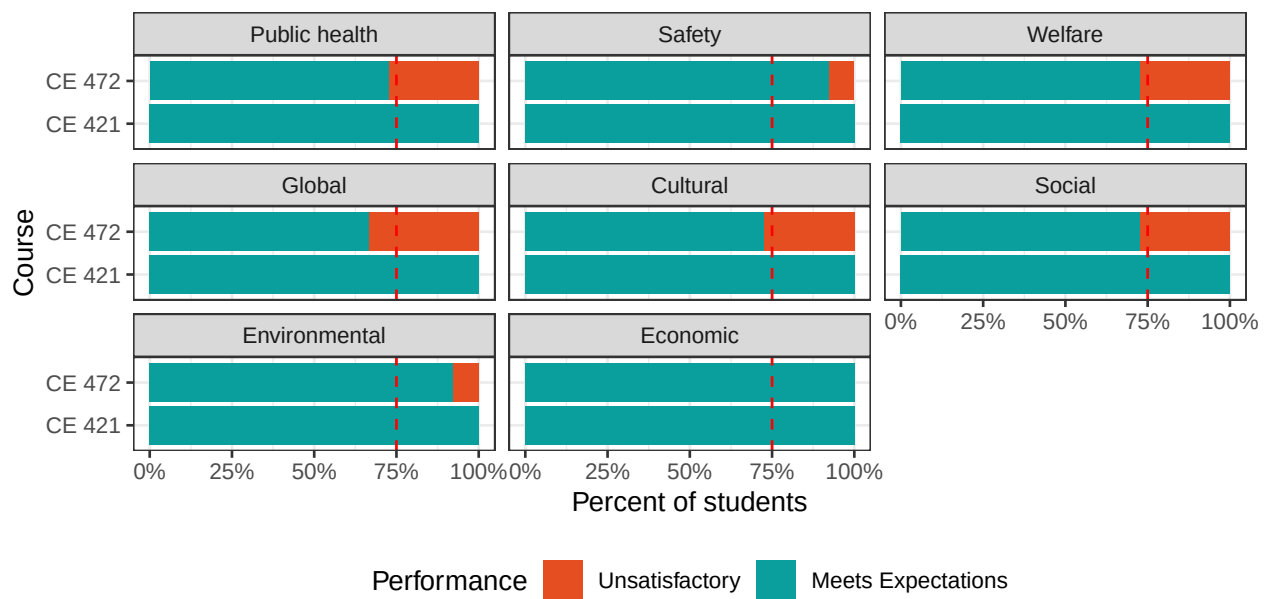


Figure 1.2: Fall 2025 and Winter 2026 Assessment of Student Outcome 2 design factors.

1.3 Communication

Students will demonstrate:

An ability to communicate effectively with a range of audiences.

1.3.1 Performance Indicators

We evaluate this outcome based on four performance indicators. Students should: 1. Write technical texts in clear and organized English 2. Apply industry conventions for professional reports (typography, reference format, etc.) 3. Use graphics to illustrate numeric data 4. Present information in oral contexts

We assess these performance indicators with the rubric in Table 1.6.

Table 1.6: Student Outcome 3 Assessment Rubric

Indicator	Unsatisfactory	Developing	Meets Expectations	Exceeds Expectations
1. Write	Unclear, disorganized, or incorrect writing interferes with understanding.	Writing has some incorrect or unclear elements.	Writing is clear and well-organized.	Writing is exceptionally clear, organized, and coherent.
2. Conventions	Does not apply industry conventions.	Applies some industry conventions.	Largely applies industry conventions.	Precisely applies industry conventions.
3. Graphics	Graphics are missing or poorly constructed.	Graphics have some construction or clarity issues.	Graphics are clear and well-made.	Graphics are exceptionally well-produced and informative.
4. Oral	Delivery is unclear or disorganized.	Delivery communicates basic ideas but lacks clarity or structure.	Delivery is clear, structured, and supports understanding.	Delivery is polished, concise, and highly effective.

1.3.2 Assessment Instruments

1.3.3 Data and Evaluation

1.4 Ethics

Students will demonstrate:

An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Performance Indicators

We evaluate this outcome based on three performance indicators. Students should:

1. Recognize personal and professional responsibilities towards others including employees, clients, the wider profession, and the public.
2. Recognize moral and ethical dilemmas in engineering situations and make judgments to identify appropriate resolutions.
3. Consider the impacts of an engineering solution in global, economic, environmental, and societal contexts.

We assess these performance indicators with the rubric in Table 1.7.

Table 1.7: Student Outcome 4 Assessment Rubric

Indica- tor	Unsatisfactory	Developing	Meets Expectations	Exceeds expectations
1. Recognize re- sponsi- bilities	Does not recognize or identify professional responsibilities.	Recognizes some professional responsibilities, but misses at least one.	Recognizes professional responsibilities towards all key stakeholders	Demonstrates interest in and motivation to serve society through the profession.
2. Make judg- ments	Does not identify ethical dilemmas, and or develops inappropriate resolutions	Misidentifies ethical dilemmas, and or develops inadequate resolutions.	Properly identifies ethical dilemmas and identifies appropriate resolutions.	Actively considers ethical challenges that may arise in professional contexts and seeks to mitigate them.
3. Con- sider im- pacts	Fails to identify impacts in any contexts.	Identifies impacts in some contexts, but fails to identify likely impacts in others.	Identifies likely impacts in a variety of contexts.	Actively seeks to consider the ethical impact of engineering situations in global, economic, environmental, and societal contexts.

Assessment Instruments

We evaluated this outcome most recently in Winter 2025. All graduating students take CE 472 in Winter semester. Students take CE 291 R each semester from their sophomore year onward. CCE 102 may be taken in Fall or Winter semesters, so this is approximately a 50% sample of the students in the program. CCE 102 also includes students who are in the CFM and the FM programs.

We assess this student outcome in the following courses with the assessments listed in Table 1.8. Complete instrument prompts are available in the program assessment folder.

Table 1.8: Assessment Instruments for Student Outcome 2

Assessment Level	Course	Instrument
Consider Impacts		
Formative	CCE 102	HW 3.4 Common Goods - Impacts tables
Summative	CE 472	Ethics Reflection
Make Judgments		
Formative	CCE 102	Ethics case study
Summative	CE 472	Ethics Quiz (Q7)
Summative	CE 472	Ethics Reflection
Recognize Responsibilities		
Formative	CCE 102	Code of Ethics Reflection
Formative	CCE 102	Code of Ethics write-up
Formative	CE 291R	Final reflection exercise, given to students to complete during final exams.
Summative	CE 472	Ethics Quiz (Q1-6)
Summative	CE 472	Ethics Reflection

1.4.1 Data and Evaluation

We evaluated this outcome most recently in Winter 2025. All graduating students take CE 472 in Winter semester. Students take CE 291 R each semester from their sophomore year onward. CCE 102 may be taken in Fall or Winter semesters, so this is approximately a 50% sample of the students in the program. CCE 102 also includes students who are in the CFM and the FM programs. Figure 1.3 shows the share of students at each performance level for each indicator for each course, alongside the threshold of 75%. By taking CE 472, 75% of students meet the expectations for each indicator.

The evaluation of this assessment data led the instructors of CCE 102 and CE 472 to make changes in the way that they taught and evaluated ethical dilemmas. The re-assessment of this outcome will occur in the 2027-2028 academic year.

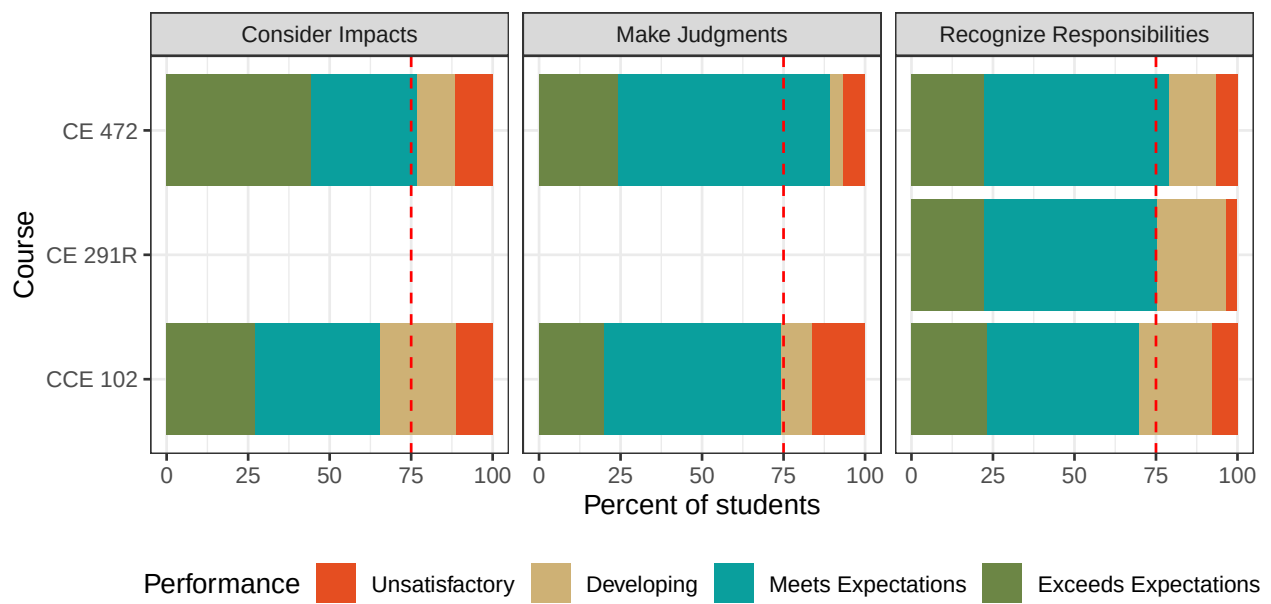


Figure 1.3: Winter 2025 Assessment of Student Outcome 4

1.5 Teamwork

Students will demonstrate:

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

1.5.1 Performance Indicators

We evaluate this outcome based on three performance indicators. Students should:

1. Show leadership through accountability and service.
2. Contribute to a collaborative and inclusive environment where all team members feel a sense of belonging.
3. Meet team objectives through establishing goals and planning tasks.

We assess these performance indicators with the rubric in Table 1.9.

Table 1.9: Rubric to assess student outcome 5: Teamwork

Indicator	Unsatisfactory	Developing	Meets Expectations	Exceeds Expectations
1. Accountability and service	Does not accomplish assigned tasks.	Accomplishes assigned tasks but is delayed or requires reminders.	Accomplishes tasks on time.	Accomplishes tasks on time, and helps team members accomplish their roles.
2. Contribute to team belonging	No or negative involvement in team activities.	Involved in team activities and makes some effort to engage others.	Actively involved in team activities and encourages engagement from others.	Drives team activities and motivates productive engagement from others.
3. Meet team objectives	Fails to identify reasonable goals or appropriate tasks.	Develops goals and tasks that are infeasible, or that required substantial revision.	Develops goals and a task plan that results in meeting the assignment expectations.	Develops goals and a resilient task plan that results in exceeding the assignment expectations.

1.5.2 Assessment Instruments

We assess this student outcome in the following courses with the assessments listed in Table 1.10. Complete instrument prompts are available in the program assessment folder.

1.5.3 Data and Evaluation

We evaluated this outcome most recently in Winter 2026. Current data include formative assessment in CCE 102 and summative assessment in CE 472. Figure 1.4 shows the share of students at each performance level for each indicator for each course, alongside the threshold of 75%. Students in CCE 102 met the threshold for all three indicators. Students in CE 472 were below the threshold for all three indicators, with 69 percent of students meeting or exceeding expectations for each indicator.

Table 1.10: Assessment Instruments for Student Outcome 5

Assessment Level	Course	Instrument
Accountability		
Formative	CCE 102	Student project team survey
Summative	CE 472	Self reflection question
Belonging		
Formative	CCE 102	Student project team survey
Summative	CE 472	Self reflection question
Objectives		
Formative	CCE 102	Student project team survey
Summative	CE 472	Self reflection

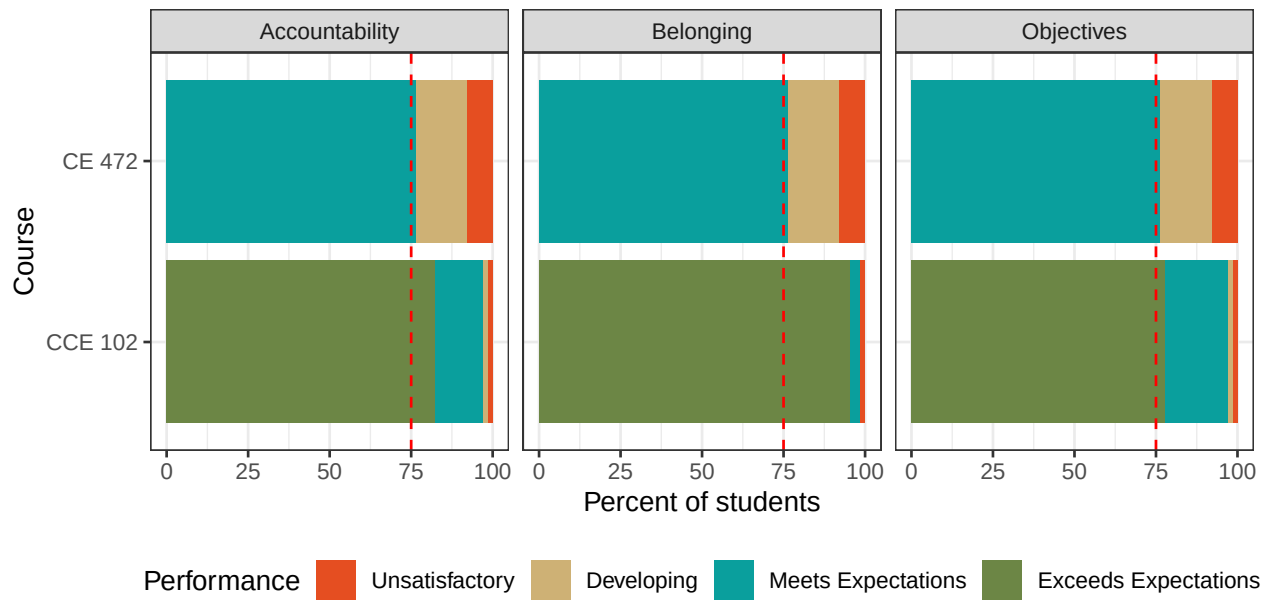


Figure 1.4: Fall 2025 and Winter 2026 Assessment of Student Outcome 5

1.6 Experiments and Data

Students will demonstrate:

An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

1.6.1 Performance Indicators

We evaluate this outcome based on four performance indicators. Students should:

1. Identify conditions: Identify what conditions need to be controlled or varied in an experiment in order to obtain data needed to test a hypothesis.
2. Select methods: Select methods, materials, and/or equipment that will generate the desired data.
3. Conduct: Conduct the experiment and keep appropriate records.
4. Analyze and conclude: Analyze data using scientific techniques and draw supported conclusions.

We assess these performance indicators with the rubric in Table 1.11.

Table 1.11: Rubric to assess student outcome 6: Experiments and Data

In- di- ca- tor	Unsatisfactory	Developing	Meets Expectations	Exceeds expectations
1. Identify conditions	Does not distinguish controls and variables; attempts multiple variables in one experiment	Identifies some controls and variables	Correctly identifies all controls and variables	Correctly identifies all controls and variables as well as possible confounding conditions
2. Select methods	Does not match experiment	Selects some adequate methods, materials, or equipment	Selects adequate methods, materials, and equipment; can justify choice	Selects adequate methods, materials, and equipment; can justify choice; may use standards
3. Conduct	Careless setup or execution of experiment; little or no recordkeeping; inattentive to experiment	Some problems in setup or execution; keeps some records	Carefully sets up and executes experiment; records include photos, notes, observations, and potential errors	Carefully sets up and executes experiment; records include photos, notes, observations, and potential errors or limitations
4. Analyze and conclude	Little or no analysis of data; incorrect analysis of data; no conclusion; concludes without data; concludes contradictory to data	Shows some analysis by figures, equations, statistics, etc., though perhaps with mistakes; may conclude without data or contradictory to data	Demonstrates thorough analysis by figures, equations, statistics, etc.; concludes based on data and in accordance with data; recognizes assumptions	Demonstrates thorough analysis by figures, equations, statistics, etc.; concludes based on data and in accordance with data; recognizes assumptions; acknowledges civil engineering applications of results

Table 1.12: Assessment Instruments for Student Outcome 6

Assessment Level	Course	Instrument
Identify Conditions		
Formative	CE 232	Bernoulli lab
Summative	CE 341	Soil lab
Summative	CE 351	Water treatment jar test
Select Methods		
Formative	CCE 270	Assessment not specified
Formative	CE 232	Bernoulli lab
Summative	CE 341	Soil lab
Conduct		
Formative	CCE 270	Assessment not specified
Formative	CE 232	Bernoulli lab
Summative	CCE 306	Soil and aggregate lab
Summative	CE 341	Soil lab
Summative	CE 351	Water treatment jar test
Analyze and Conclude		
Formative	CE 232	Bernoulli lab
Summative	CCE 306	Soil and aggregate lab
Summative	CE 341	Soil lab
Summative	CE 351	Water treatment jar test

1.6.2 Assessment Instruments

We assess this student outcome at three levels:

- CCE 270: Computer Methods (introductory)
- CE 232: Fluid Mechanics (reinforcing)
- CE Breadth classes (evaluation): CCE 306 Civil Engineering Materials, CE 341 Soil Mechanics, and CE 351 Environmental Engineering

All CE students must take one of the three breadth classes where we assess this outcome, though many will take more than one.

We assess this student outcome in the following courses with the assessments listed in Table 1.12.

1.6.3 Data and Evaluation

The current data include Winter 2025 assessments in CE 232 and the breadth courses and Winter 2026 assessment in CCE 270. Figure 1.5 shows the share of students assessed at each performance level for each indicator in the relevant courses. The CE 232 formative assessment meets the threshold for all four indicators. The CCE 270 formative assessments are just below the threshold for Select Methods and Conduct. Summative breadth-course evidence is uneven: CCE 306 and CE 341 meet the threshold for Conduct, CE 341 meets the threshold for Identify Conditions, and the remaining summative assessments are below the threshold.

The committee recommends calibration among the breadth-course laboratory instructors, with particular

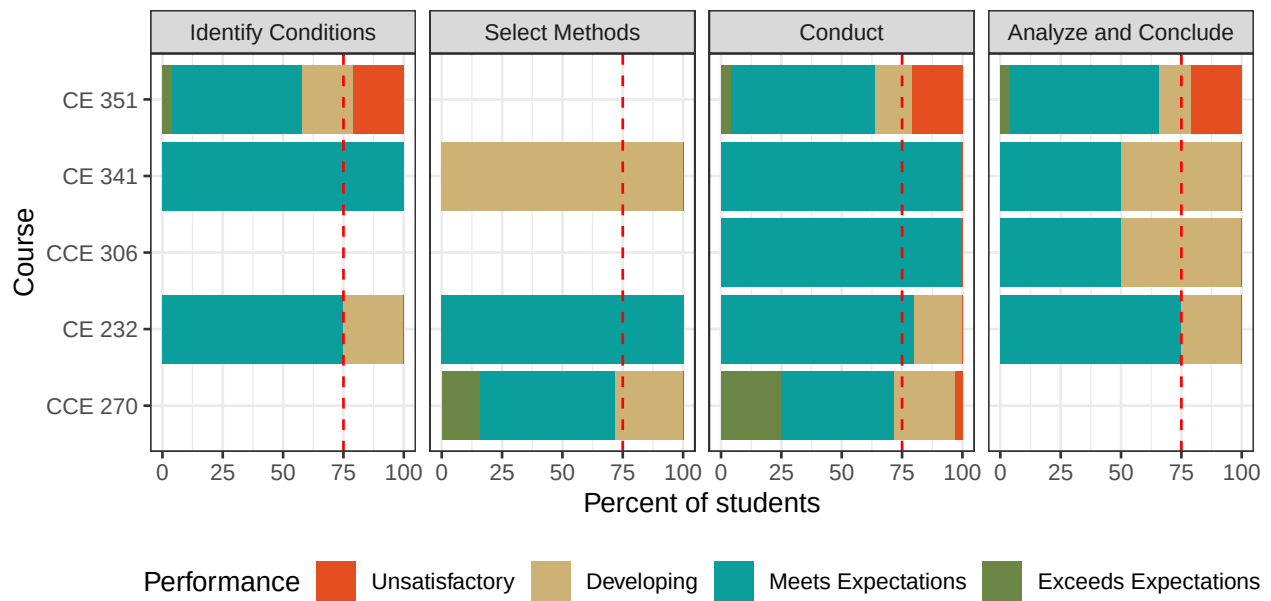


Figure 1.5: Winter 2025 and Winter 2026 Assessment of Student Outcome 6

attention to Select Methods and Analyze and Conclude, so that expectations and scoring are consistent across the courses used for summative assessment.

1.7 Acquire Knowledge

Students will demonstrate:

An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

1.7.1 Performance Indicators

We evaluate this outcome based on three performance indicators. Students should:

1. Effectively seek out appropriate resources to acquire knowledge and understanding of unfamiliar concepts.
2. Critically evaluate sources for reputability and reliability.
3. Appropriately apply newly acquired knowledge and skills to complete an analysis.

We assess these performance indicators with the rubric in Table 1.13.

Table 1.13: Student Outcome 7 Assessment Rubric

Indicator	Unsatisfactory	Developing	Meets Expectations	Exceeds Expectations
1. Seek resources	Relies exclusively on provided materials; shows little or no initiative in finding new resources.	Consults additional resources when prompted and demonstrates partial understanding of how the resources contribute to learning.	Independently identifies and uses credible, relevant resources that contribute meaningfully to understanding.	Seeks diverse, high-quality resources and synthesizes information from multiple sources to deepen understanding.
2. Evaluate sources	Resources used, if any, are irrelevant, unreliable, or misinterpreted.	Uses resources with limited evaluation of quality or relevance.	Provides evidence of evaluating resource quality and applicability.	Critically analyzes the strengths and limitations of sources.
3. Apply knowledge	Fails to apply newly acquired techniques, or applies them incorrectly, demonstrating misunderstanding or lack of effort to learn.	Attempts to use new techniques but with significant errors or incomplete understanding; requires guidance to complete the analysis.	Correctly applies newly learned techniques and practices to perform an appropriate analysis with minimal errors.	Applies new techniques accurately and efficiently, extending them beyond the basic requirements.

1.7.2 Assessment Instruments

We assess this student outcome in the following courses with the assessments listed in Table 1.14. Complete instrument prompts are available in the program assessment folder.

1.7.3 Data and Evaluation

We evaluated this outcome most recently in Winter 2026. Current data include formative assessment in CCE 113 and CCE 114 and summative assessment in CE 471. Figure 1.6 shows the share of students at each

Table 1.14: Assessment Instruments for Student Outcome 7

Assessment Level	Course	Instrument
Seek Resources		
Formative	CCE 113	Semester Project Presentation
Summative	CE 471	Research paper (F25)
Summative	CE 471	Research paper (W26)
Evaluate Sources		
Summative	CE 471	Research Paper W26
Summative	CE 471	Research Project F25
Apply Knowledge		
Formative	CCE 113	Final project
Formative	CCE 114	Practical final exam

performance level for each indicator for each course, alongside the threshold of 75%. The CE 471 summative assessment meets the threshold for Seek Resources and Evaluate Sources. The current data file does not include a nonzero summative assessment for Apply Knowledge; formative assessment of Apply Knowledge in CCE 113 and CCE 114 is slightly below the threshold.

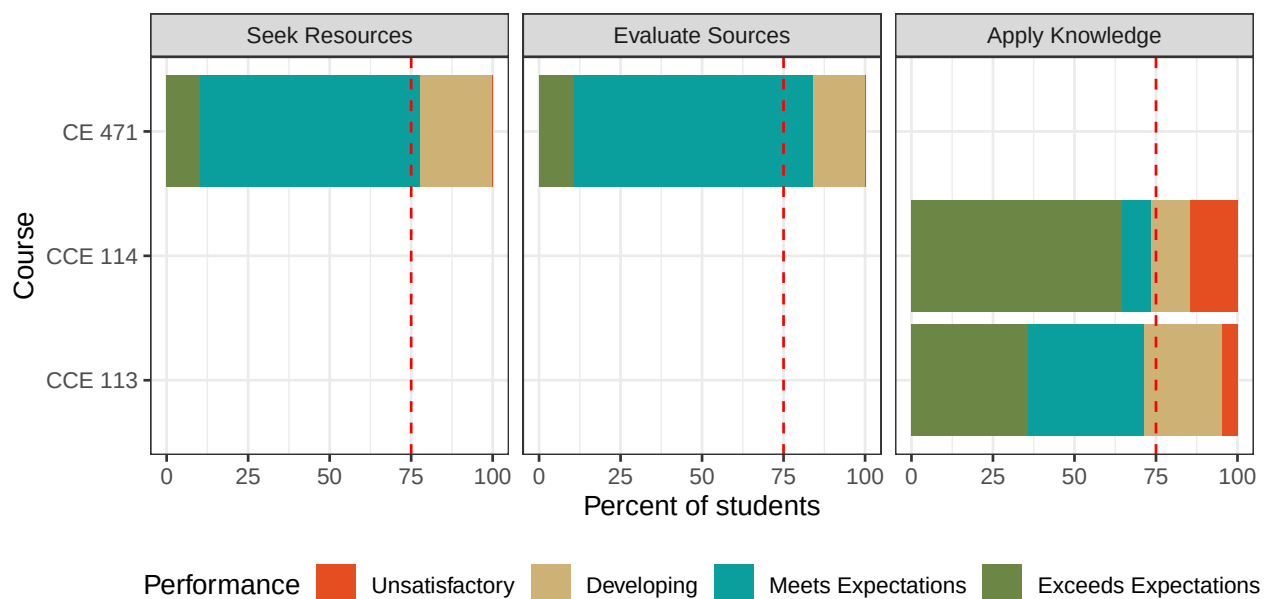


Figure 1.6: Fall 2025 and Winter 2026 Assessment of Student Outcome 7

The committee recommends collecting complete summative evidence for all three Acquire Knowledge indicators in the next assessment cycle, with particular attention to Apply Knowledge.

Chapter 2

Exit Surveys and Interviews

2.1 Exit Surveys

Each December and April, the program sends a survey to all of our graduating students. The purpose of this survey is to collect data on the following items:

1. Self-assessment of student outcomes
2. Prospective employment or graduate school plans
3. Belonging and extracurricular involvement
4. Program Evaluation

In Winter 2026, the committee streamlined and overhauled the survey to give more targeted answers and avoid duplication with surveys administered by BYU Alumni. 21 students responded to the 2026 survey.

2.1.1 Self-assessment of student outcomes

The survey asks students to self-assess their preparedness on the student outcomes and related preparation categories described in Chapter 1 along a five-point scale from “Not at all prepared” to “Very well prepared.” Two outcomes are divided into separate categories: SO3 Communication is divided into written and oral communication, and SO2 Design is divided into design solutions and design impacts (as in the “consider factors” indicator).

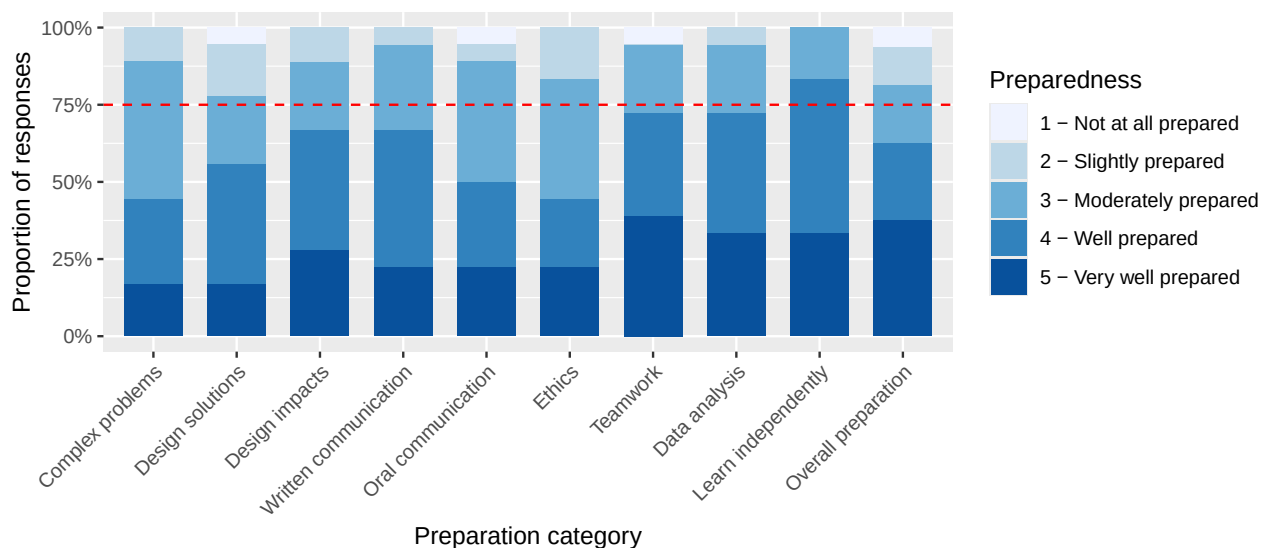


Figure 2.1: Student self-assessment response proportions by preparation category

In only one category (Learn independently) do at least 75% of students rate their preparation as “Well prepared” or higher. The categories with the most potential for growth in this area are Complex Problems and Ethics.

2.1.2 Prospective employment or graduate school plans

The survey asks students to report their immediate plans after graduation and, where applicable, their future graduate school plans. Figure 2.2 shows the distribution students across these levels. More than half of the students suggest that they will pursue some form of graduate education in their future.

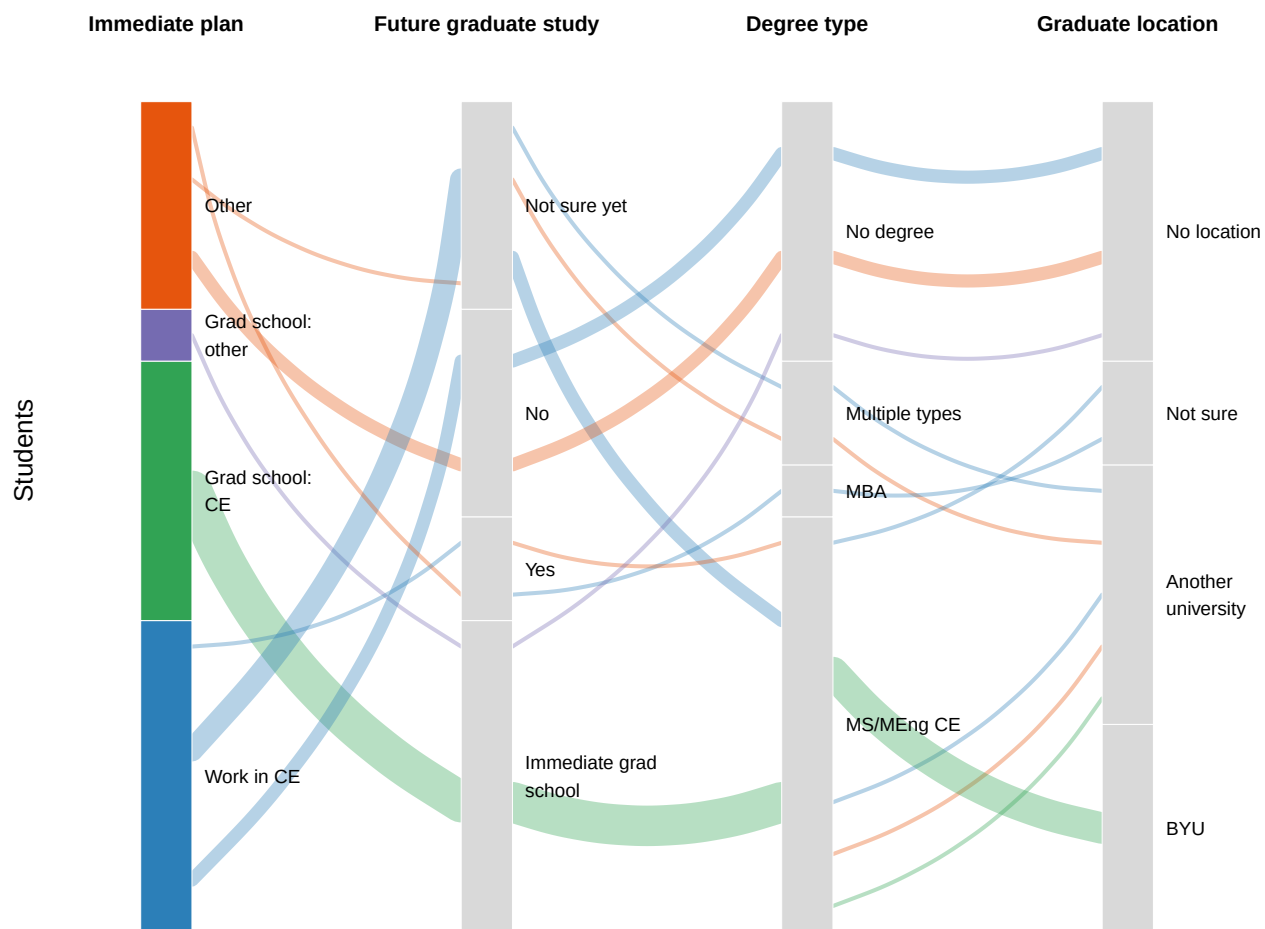


Figure 2.2: Prospective employment and graduate school plans

2.1.3 Belonging and extracurricular involvement

The survey asks students to evaluate several elements of belonging and to report the number of meaningful learning experiences they completed during their time at BYU.

Figure 2.3 shows the distribution of students' responses to the belonging questions. Student generally report strong levels of agreement with the relevant statements. There are three statements however, where less than 75% percent of students report "Agree" or "Strongly agree":

2. I feel a connection to the Civil & Construction Engineering department.
3. I feel a connection to the College of Engineering.
4. The Civil Engineering program provided meaningful opportunities for involvement outside the classroom.

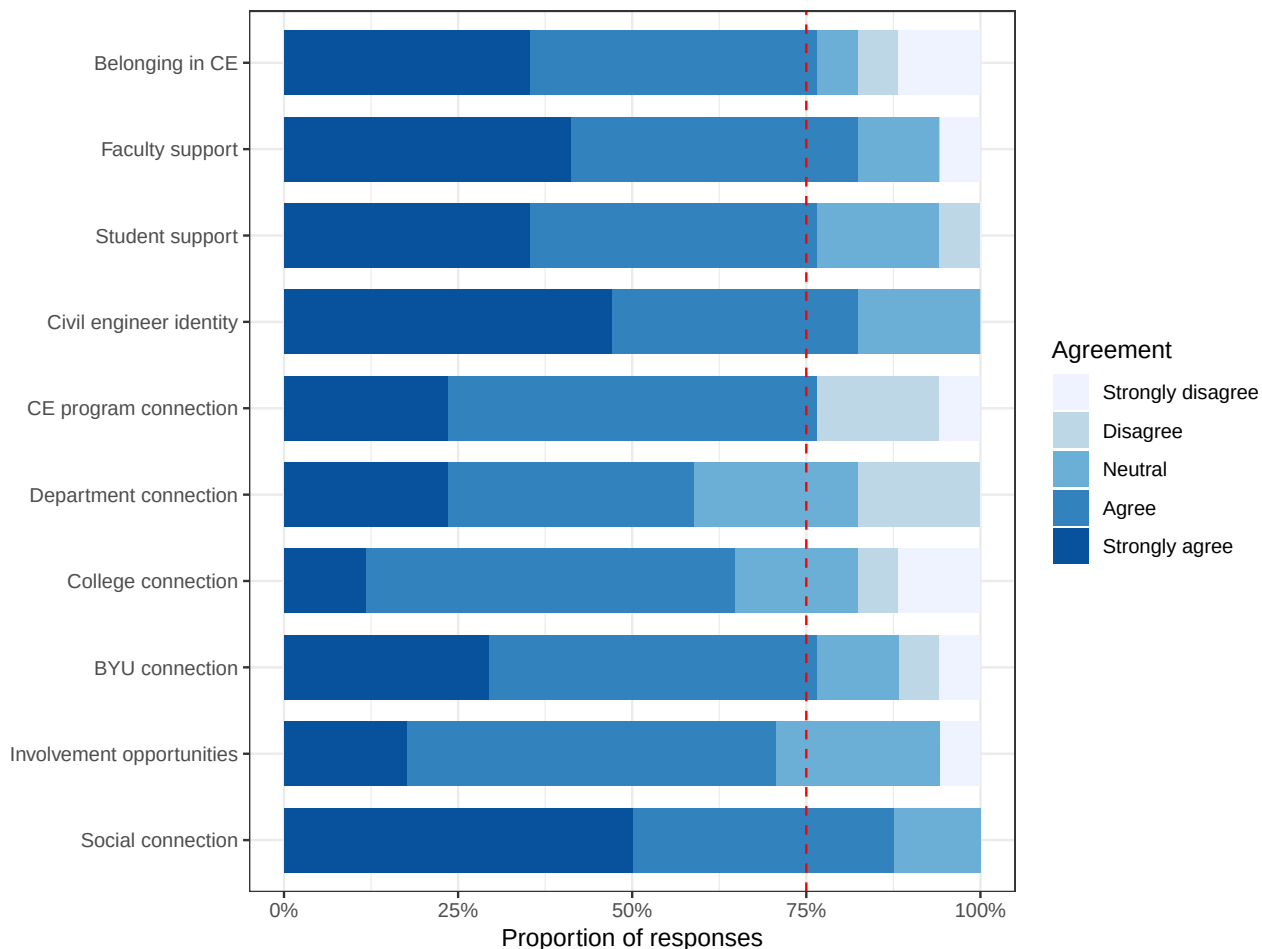


Figure 2.3: Student agreement with belonging statements

The university has a goal for students to complete at least two learning experiences during their time at BYU. Figure 2.4 shows the distribution of the number of learning experiences reported by students. Completing the capstone is a requirement of graduation, though some students did not mark having experienced this. These are the same records as the students who report no learning experiences. The most common number of experiences is 7 or more.

The survey instructs students to count each experience once, even if it lasted multiple semesters or involved multiple activities. The question describes the experience categories as follows:

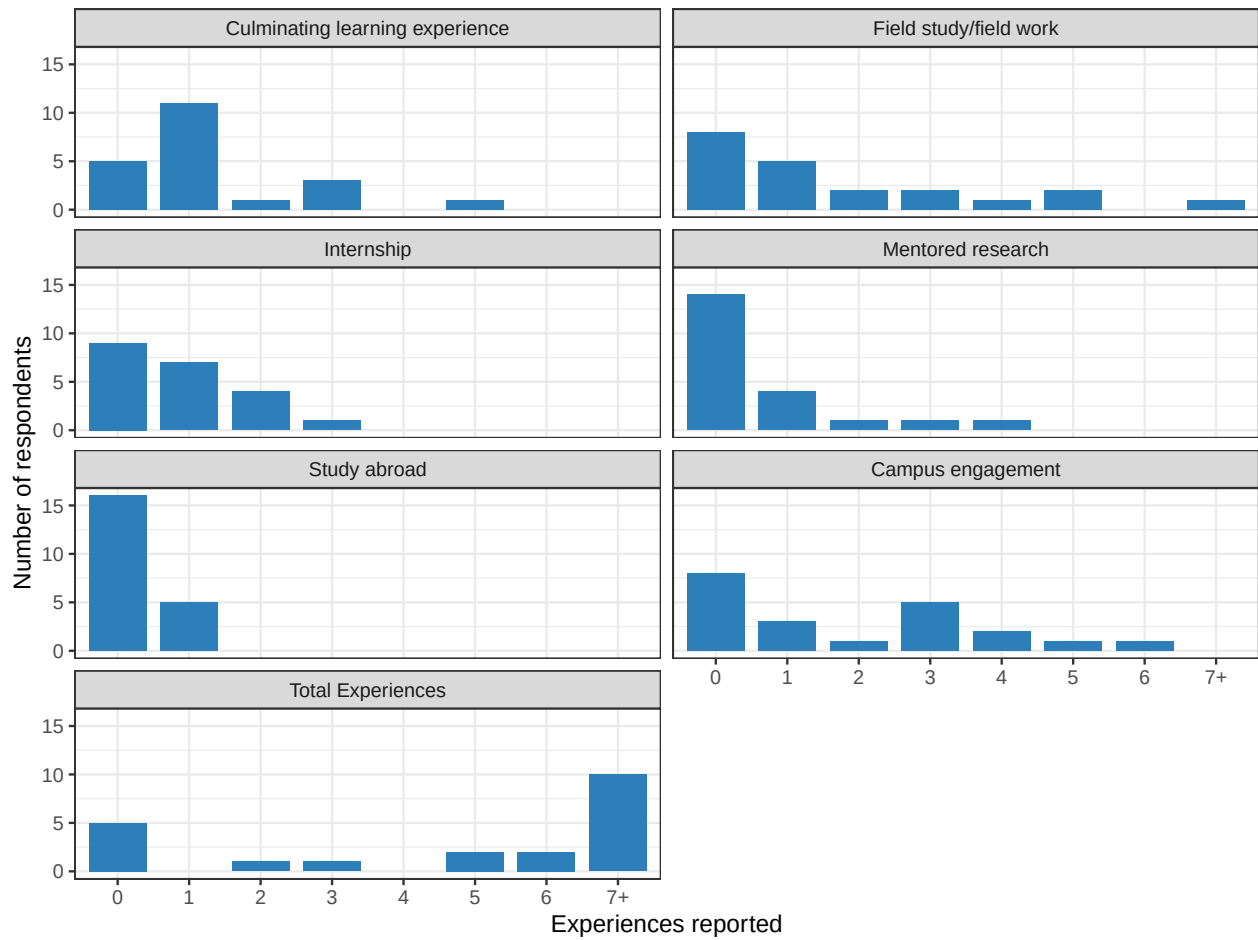


Figure 2.4: Distribution of learning experiences reported by type

- Culminating learning experience: senior capstone, design project, thesis, recital, or other major culminating academic experience.
- Field study/field work: learning or work conducted in real-world settings outside the classroom, such as site visits or field-based projects.
- Internship: structured work experience in a professional engineering or related setting.
- Mentored research: faculty-mentored research, conference presentation, or publishing work.
- Study abroad: academic experience in another country or culture.
- Campus engagement: student organizations, clubs, competitions, leadership, department activities, or other meaningful involvement outside the classroom.

2.1.4 Program Evaluation

The survey asks students to identify the CE or CCE course that contributed most to their learning and the course that contributed least to their learning. Because these are open-ended responses, Figure 2.5 counts course mentions using course numbers and common course-name phrases. Responses that identify more than one course contribute one mention to each course.

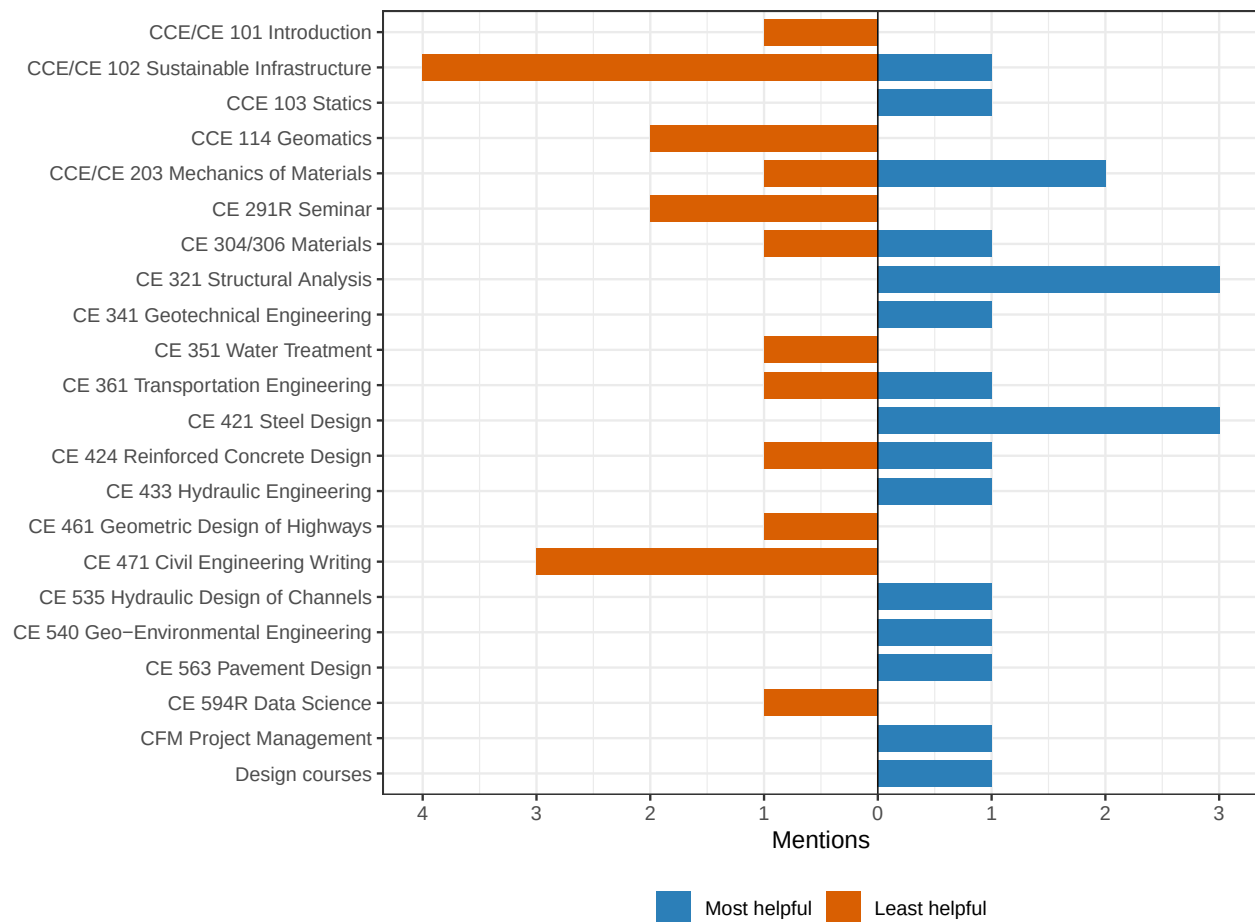


Figure 2.5: Courses mentioned as contributing most or least to student learning

Sustainable Infrastructure and Civil Engineering Writing are the two courses with the most negative mentions, which is unsurprising as these are new courses and under active development.

2.2 Exit Interviews

Each April, the undergraduate committee samples graduating students and interviews them in personal conversation. The purpose of this interview is to qualitatively assess student-observed program strengths and weakness.

The CE undergraduate coordinator randomly assigns three to four graduating seniors to each member of the committee. The committee members arrange to speak privately with each student for fifteen to twenty minutes. Students may elect to not schedule the interview. The student is invited to direct the discussion, but the committee member asks guiding questions such as:

- What are the most positive learning experiences you have had?
- Have you had any less successful learning experiences?

Each committee member makes notes of the student's responses. The assessment coordinator consolidates these notes into a coherent statement with the aid of a Large Language Model and posts it in the space below.

2.2.1 Class of 2026

The civil engineering program demonstrates several clear strengths, particularly in the quality of instruction in upper-division courses and the accessibility of faculty. Students consistently report positive experiences with professors who are engaged, approachable, and effective at connecting course material to professional practice. This is especially evident in areas such as structures, geotechnical engineering, and transportation, where well-organized courses and strong instruction enhance learning outcomes. The program also supports successful career placement, with graduates securing relevant employment, internships, and graduate school opportunities. High-impact experiences such as capstone projects, undergraduate research, and study abroad further contribute to student development. Additionally, the program fosters a strong sense of community among students, particularly in later years, where stable cohorts and peer interaction enhance the overall educational experience.

Despite these strengths, the program exhibits several systemic weaknesses related to coherence, structure, and student engagement. The introductory experience is inconsistent and, in some cases, ineffective at orienting students to the discipline or fostering a sense of belonging, particularly for transfer students. There are also notable gaps between foundational coursework and upper-division applications, including insufficient alignment between mathematics preparation and engineering problem-solving. Students report variability in course organization and grading practices, as well as occasional overemphasis on tools (e.g., programming) at the expense of conceptual understanding. The seminar program is widely viewed as inefficient in its current form, with diminishing value in upper-division years and a need for greater emphasis on career preparation and interactive engagement. Similarly, the capstone experience lacks clarity and alignment with prior coursework, and is perceived by some students as placing undue focus on low-value tasks. Finally, student participation in co-curricular activities is inconsistent, suggesting that current approaches to engagement may be insufficiently effective.

2.2.2 Class of 2025

Students described the civil engineering program as offering strong overall preparation, with a solid breadth of coursework and particularly effective introductory classes that sparked interest in the major. Many felt well prepared for the FE exam and appreciated practical, real-world applications in courses like CE 471, as well as hands-on experiences through competition teams, research, study abroad, and TA roles. Faculty were consistently viewed as enthusiastic, accessible, and invested in student success, especially in smaller upper-division classes where interaction was more direct. The program fostered a strong sense of belonging through clubs, study groups, and shared spaces, supported by active (though sometimes inconsistent) faculty involvement. Students valued flexible curriculum elements like the junior core options, and many appreciated

opportunities to integrate faith meaningfully into their education. Physical spaces such as labs and common areas also contributed positively to collaboration and community.

At the same time, students identified several persistent challenges. Seminar was widely seen as repetitive and inefficient, with too many guest speakers and not enough practical skill development, and its scheduling sometimes conflicted with internships. There were concerns about uneven course quality, disruptions from instructor turnover, limited TA support in upper-division courses, and heavy workloads that restricted extracurricular involvement. Technical skills like CAD, Revit, and Excel were not consistently reinforced, leading to skill loss over time. Writing requirements—particularly ENGL 316—were viewed as overly burdensome and less relevant than discipline-specific alternatives like CE 471, with some students suggesting earlier placement in the curriculum. Capstone grading was sometimes perceived as overly nitpicky, and some assignments felt like busywork. Students also noted scheduling limitations (e.g., course conflicts, weaker winter offerings), a lack of advanced or graduate-level electives, and a need for clearer guidance on electives and career pathways. Additional suggestions included improving inclusivity and representation (especially for women), expanding interdisciplinary and coding opportunities, increasing faculty engagement in student organizations, and addressing administrative friction points such as curriculum changes and the graduate application process.

Chapter 3

Fundamentals of Engineering

The Fundamentals of Engineering (FE) examination provides important external evidence of our program’s effectiveness in training competent civil engineering professionals. University policies prohibit us from requiring students to take the exam before graduation, but we strongly recommend that our students do so and offer an FE preparation course to help them. Table 3.1 shows the number of students and recent graduates who attempted the FE in each of the last few years, as well as the number of BS degrees awarded by the program in each year. We do not have complete FE exam data for 2022, leading to the apparent low number of attempts in that year.

The share of students passing the FE exam is shown in Figure 3.1. BYU students pass the exam at a rate considerably higher than other ABET-accredited civil engineering programs. This data is for students currently enrolled in the program, or graduating in the 12 months prior to taking the exam. We do not include alumni who take the FE more than one year after graduation in this assessment.

We also use the FE exam to assess discipline and topic subscores. Figure 3.2 provides an interactive graphic where you can choose an FE exam category and compare BYU civil engineering students and recent alumni against an “ABET comparator” group of similar institutions. BYU students generally exceed the ABET comparator group mean in all categories for almost all periods, but fall within a single standard deviation of that mean. There are a few exceptions to this performance, with BYU students modestly below the ABET comparator mean in some categories for some periods. These include the “Ethics and Professional Practice” and the “Construction Engineering” categories.

Assessment Practices

NCEES makes aggregated FE scores available to colleges for assessment purposes. The College of Engineering at BYU has a login name to this system, which is shared with the department academic

Table 3.1: FE Exam Attempts

Year	Taken FE	BS Awarded	Pct Attempting
2022	12	77	16
2023	58	70	83
2024	66	76	87
2025	51	61	84

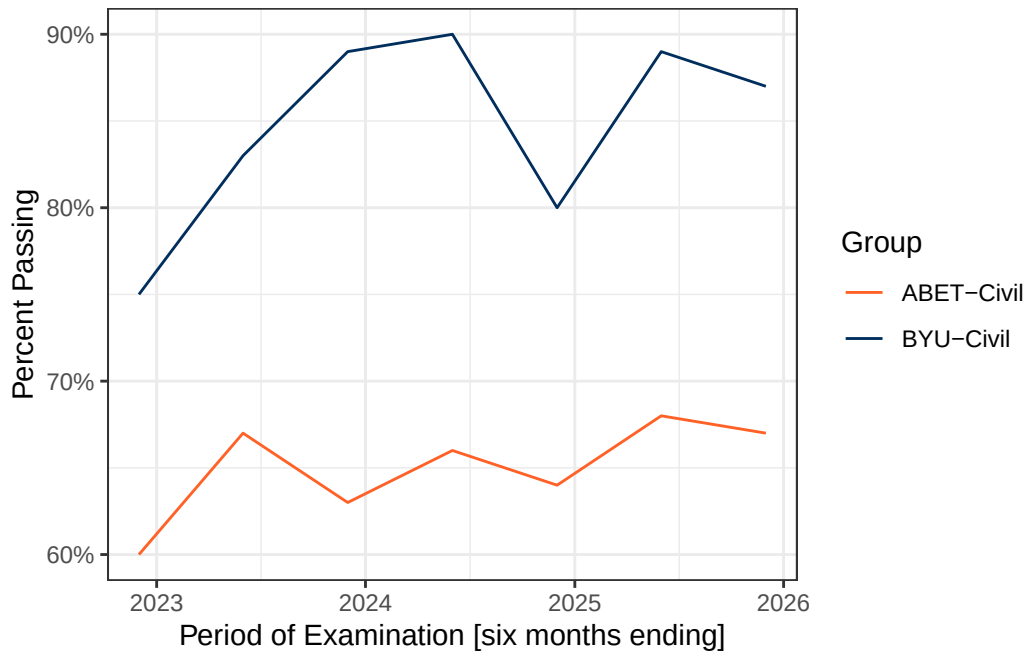


Figure 3.1: FE Exam pass rate by semester, BYU Civil and ABET comparator group.

advisor.

Each August and each February, the assessment coordinator retrieves the most recent scores from the advisor and adds them to data records in this repository. There are two places where these data are stored:

- The `data/fe.xlsx` spreadsheet contains the data from the advisor's login as well as a worksheet of the general exam pass rates.
- The `apps/data.csv` files contains the exam subscores in machine-readable format.

To update the report, the assessment coordinator does the following:

1. Runs the chunks on this page to update the FE pass rate figure.
2. Generates the shiny app in the `apps` folder with the new data.
3. Uploads the shiny app to the posit connect cloud and updates the links to the app in the report.
4. Edits the report text to reflect any additional evaluation of the assessment data.

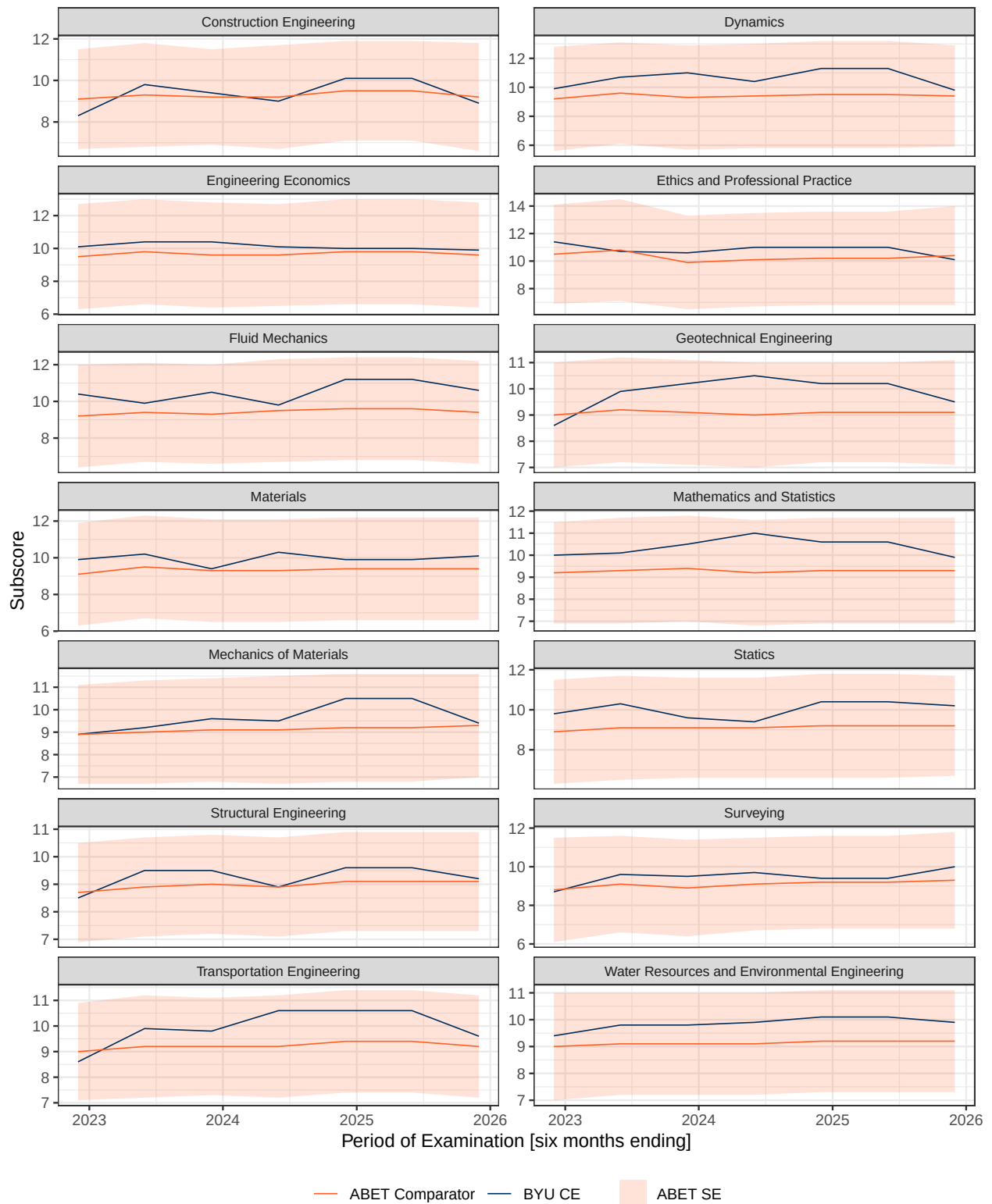


Figure 3.2: FE Exam category subscores from Fall 2022 onwards. Interactive application on ShinyApps.

Chapter 4

Student Course Evaluations

In completing the evaluation of each course, students indicate how well they felt the course advanced the Aims of a BYU Education, which are that a BYU Education should be

- Spiritually strengthening
- Intellectually enlarging
- Character building
- Leading to lifelong learning and service

Students rate each course on a five-point scale for each Aim. Figure 4.1 shows the Department averages for each course over the last few years. These evaluations have remained largely stable across all categories, but these numbers unfortunately cannot be disaggregated by program.

Assessment Practices

The program coordinator has access to a dashboard at data.byu.edu that presents the aggregated course evaluation data.

Each May, the assessment coordinator retrieves the most recent course evaluation data from the portal and updates the image in the `data/student_ratings.png` file and renders the page.

The students also respond to questions about course organization, the helpfulness of the instructor and materials, and other related topics. The responses to these questions help to identify instructor and course strengths and weaknesses, which inform individual faculty development as well as status and rank advancement decisions. This information is not used in program assessment.

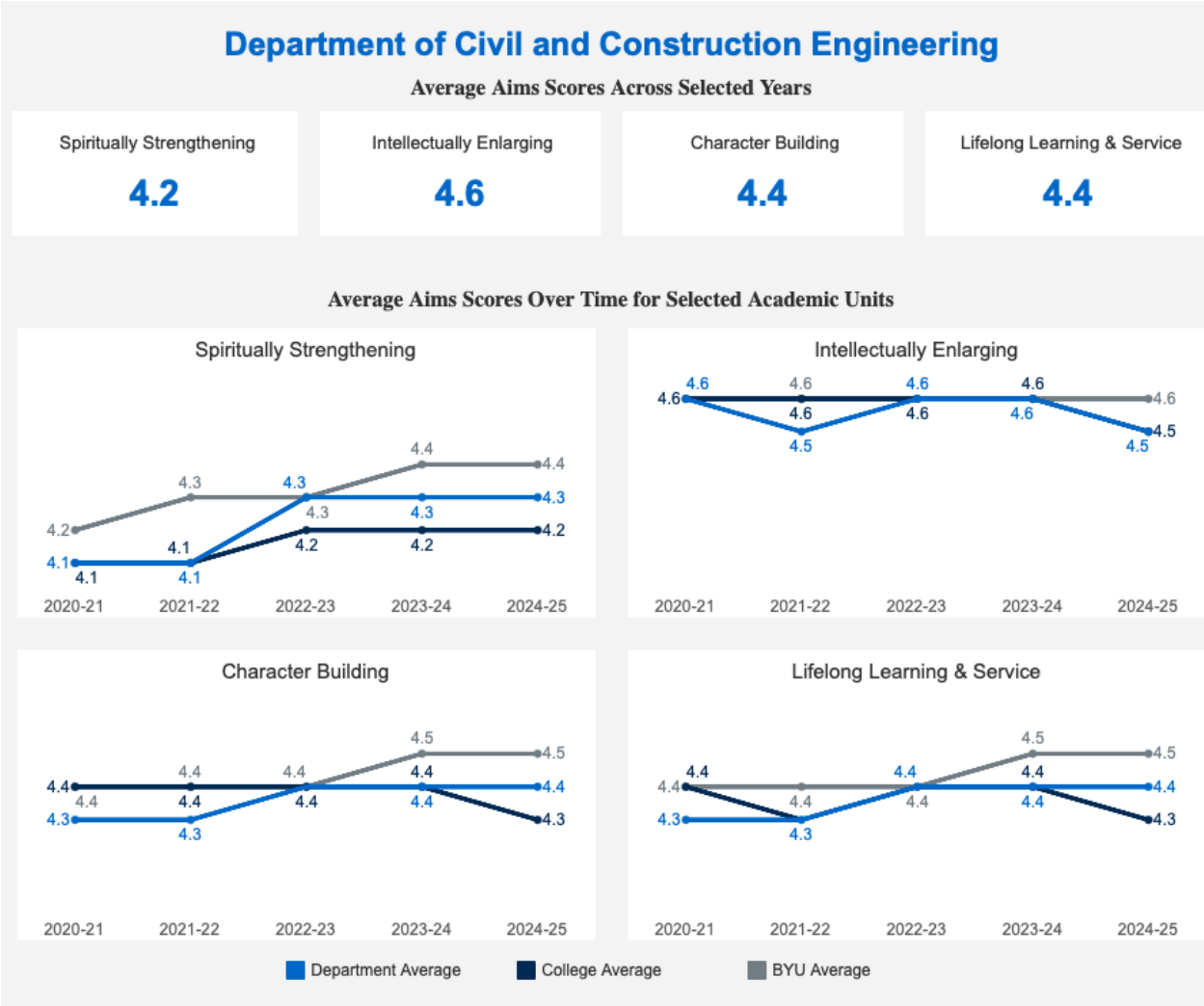


Figure 4.1: Department course evaluation averages.

Chapter 5

Student Progress

The CE program is interested in ensuring that students are progressing in the major. We do this by advocating for students to meet regularly with the department academic advisor; monitoring courses with high rates of D, E, or W grades; and tracking student time to graduation.

5.1 Student Advisement

Student advisement is an important part of student progress in the major. The department academic advisor helps students understand program requirements, sequence prerequisites, plan for graduation, and resolve registration barriers. The advisor also provides early guidance to prospective students, new admits, transfer students, and students returning from leave or mission deferment.

The university requires students to meet with the department academic advisor to clear for graduation. The university also requires students to meet with the department academic advisor to register for courses after being placed on academic probation to develop an academic improvement plan. Beyond these required meetings, faculty encourage students to meet with the advisor early to discuss course selection and major requirements.

The program uses advisement information as contextual evidence when evaluating student progress statistics. Advisement patterns can help the undergraduate committee identify barriers in the curriculum, confusing prerequisite sequences, course availability problems, or points where students need clearer information. These data are not used to assess individual student performance.

Each year, the undergraduate committee should review aggregate advisement statistics with the department academic advisor. Relevant statistics may include the number of advising appointments, the number of unique students advised, appointment timing during the academic year, common advising topics, and the number of graduation plans reviewed. When available, the advisor should also identify recurring issues that affect student progress, such as bottleneck courses, transfer-credit questions, prerequisite problems, or registration constraints.

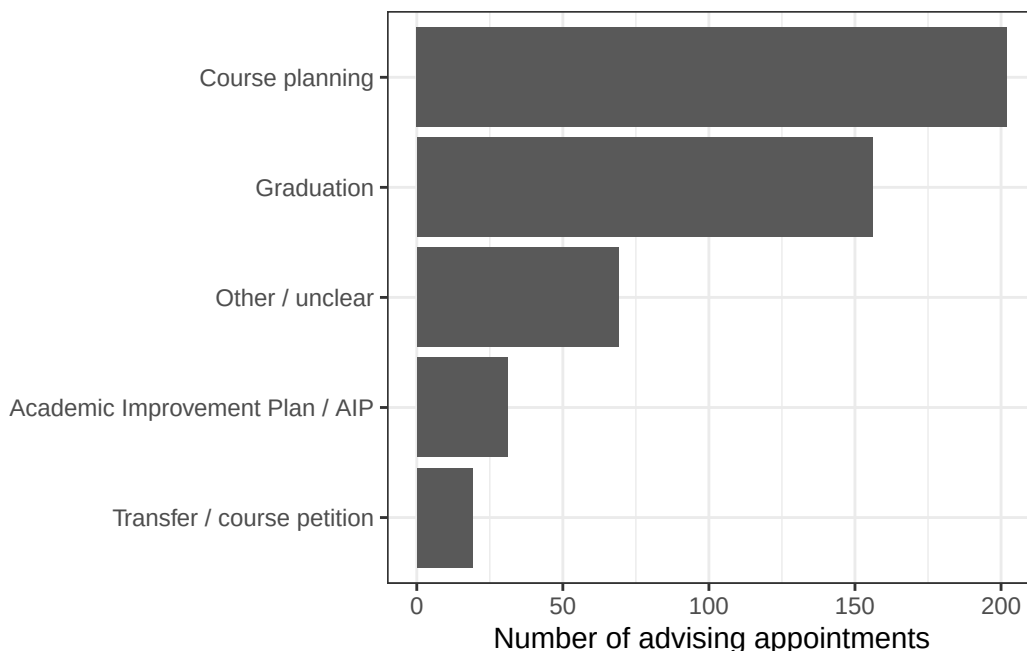


Figure 5.1: Number of advisement appointments by primary topic for CE students from May 2025 to May 2026.

From May 2025 to May 2026, the department advisor held 477 appointments with enrolled CE majors. Figure 5.1 shows the number of advisement appointments by the primary topic identified from the student's appointment question. It should be noted that this data only include students who registered for an appointment with the advisor; students who came to open hours without an appointment are not included.

Assessment Practices

The department academic advisor provides an annual summary of student advisement activity to the undergraduate committee each summer. The summary should report aggregate statistics only and should not include personally identifying student information.

The undergraduate committee should use the advisement summary alongside the time-to-graduation and DEW-rate data when evaluating student progress in the major. If advisement data point to recurring curricular or administrative barriers, the committee should document the issue and any recommended action in the annual program stewardship report.

5.2 DEW Rates

The university has asked us to track courses and students that earn a D, E, or W grade. We are developing tools to track this information.

5.3 Time to graduation

The University tracks time to graduation for each student in number of enrollments, or the number of semesters and terms that a student enrolls in courses. This controls for mission deferments but not for transfers from other institutions. The university target is for 75 percent of students to graduate within twelve enrollments, though the university total of 61 percent is less than this target.

Figure 5.2 shows the share of CE program students who graduate within the specified number of enrollments. This includes all students who entered the university in the 2017-18 through 2021-22 school years, meaning a cohort of five years of entering students, with the most recent students in the cohort entering four complete academic years prior to the present.

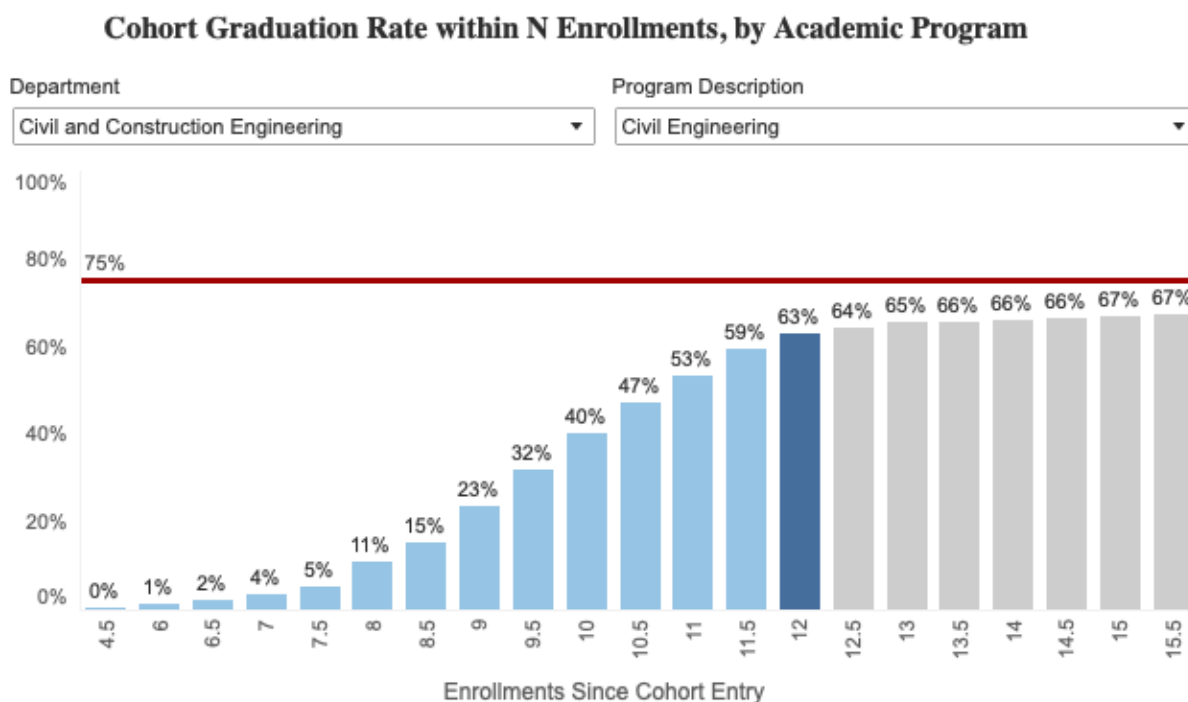


Figure 5.2: Enrollments to graduation for CE program students. Cohort entering BYU from 2017-18 through 2021-22 school years.

For this period, 273 of 435 total students graduated within twelve enrollments, for a share of **63 percent**. The time to graduation for each student is not given to us by the university. But we can track the aggregate statistic over time. Figure 5.3 shows the share of CE program students who graduate within twelve enrollments for rolling five year aggregated cohorts. The most recent 63% rate is a modest increase over the previous year, but there is an overall negative trend.

Warning

It should be noted that if a student entered the university in 2020-21 and only enrolls in fall and winter semesters, they would have only completed ten enrollments by the end of the 2024-25 academic year.

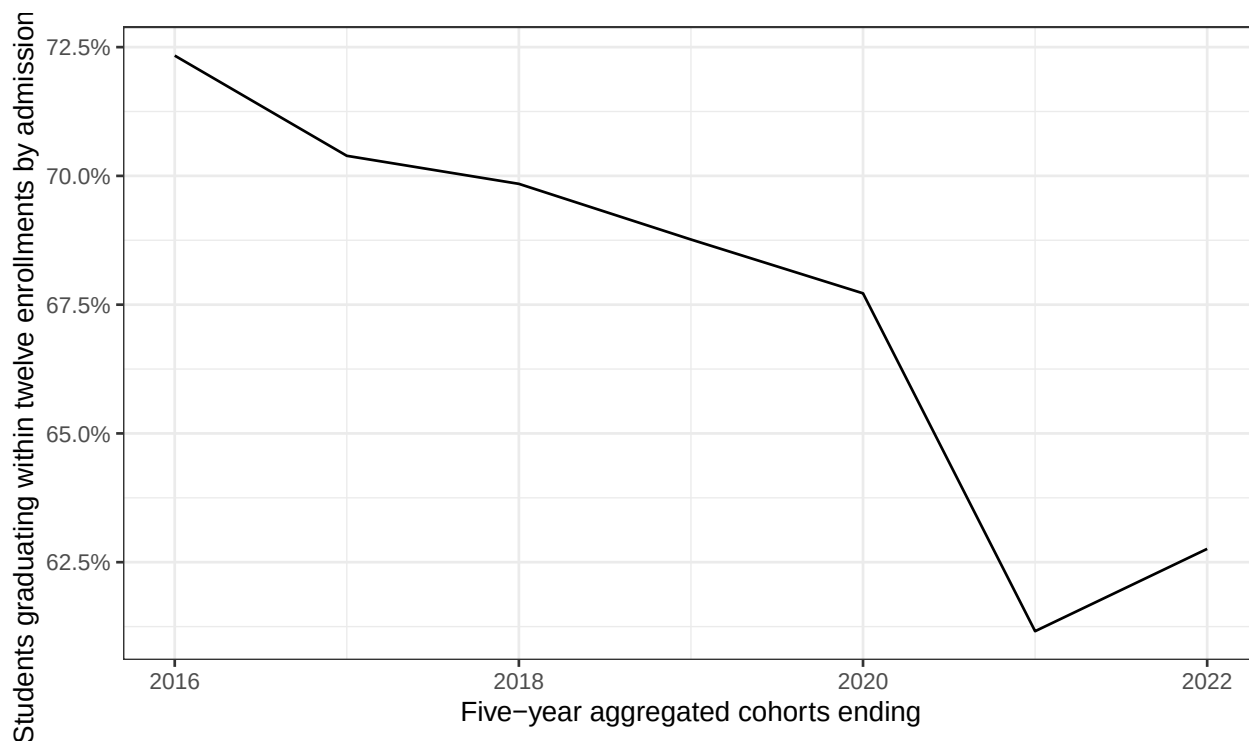


Figure 5.3: Percent of CE students graduating within twelve enrollments, five year rolling enrollment cohorts.

i Assessment Practices

The program coordinator has access to a dashboard at data.byu.edu that presents the time to graduation data. The dashboard is located at the above website under **Student Success Indicators > Enrollments to Graduation**. The resulting dashboard allows the user to filter to specific programs and other student types.

New data should be added to this report each June. The program assessment coordinator should obtain the new data and update this page.

The image showing the cohort progression should be for five years of students, ending four academic years ago. The image should be saved in the **data/graduation** folder and the link updated in this report. The numeric information beneath this image should be added to the table in **data/graduation/cohorts_in_12.xlsx**, and the page should be rendered.